

# **HAND TRACKING**

**Project Report**

**Submitted to**



"ज्ञानदेव तु कै वल्यम"

**Shri Shankaracharya Technical Campus, Bhilai**

**Affiliated To**

**Chattisgarh Swami Vivekanand Technical University**

**In Partial Fulfillment of Vocational Training**

**In**

**Artificial Intelligence**

**By**

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**Under the guidance of : Archana Mam**

## **ABSTRACT**

This project, **Hand Tracking**, demonstrates the use of computer vision and artificial intelligence to detect and track human hand movements in real time using a standard webcam. The objective of the project is to develop a touch-free human-computer interaction system capable of accurately identifying hand landmarks and utilizing them for gesture-based applications.

The system is implemented in Python using OpenCV and MediaPipe. OpenCV is used for capturing and processing video frames, while MediaPipe provides a robust hand-tracking framework capable of detecting and tracking 21 hand landmarks in real time. The tracked landmarks are processed to determine the position and movement of the hand, enabling gesture recognition and real-time interaction. The application provides visual feedback by displaying the detected landmarks and their connections, ensuring accurate and responsive tracking.

The project demonstrates the effectiveness of modern computer vision techniques in achieving reliable real-time hand tracking without the need for specialized sensors or wearable devices. Its low hardware requirements and efficient performance make it suitable for a wide range of applications, including gesture-controlled interfaces, virtual and augmented reality, robotics, gaming, sign language recognition, and assistive technologies.

Overall, the project provides a practical introduction to real-time hand tracking and highlights its potential as a foundation for developing advanced gesture-based human-computer interaction systems.

## **DECLARATION BY THE CANDIDATE**

I hereby declare that the vocational training report entitled “HAND TRACKING” is based on my own work carried out during the training period at Techonet, Bhilai, under the guidance of Mrs.Archana Mam

I assert the statements made and conclusions drawn are an outcome of my research and training work. The work contained in the report is original and has been done by me under the general supervision of my supervisor. Also, the work has not been submitted to any other Institution for any other degree/diploma/certificate in this university or any other University of India or abroad. I have followed the guidelines provided by the university in writing the report. Whenever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them in the text of the report and giving their details in the references.

**Name of the candidate: APURVA NAYAK**

**Signature:**

**Roll no.: 301402223029**

## **ACKNOWLEDGEMENT**

I take this opportunity to extend my sincere gratitude to Techonet, Bhilai, for providing me with an incredible platform to gain hands-on experience in the cutting-edge fields of the implementation of the Python and the integration of data science libraries such as scikit-learn, NLTK, and NumPy. This training was a significant step in bridging the gap between theoretical knowledge and practical software application.

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